MIT Academy of Engineering

EDS Assignment 2:- Data Analysis & Scripting

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Batch:-	CS3 [C-31]
Dataset:-	Amazon Product Sales & Reviews (2023)

Dataset Loading Code

I am using the code typed below to accessing the dataset.

```
import pandas as pd
import numpy as np

# Load the dataset
df = pd.read_csv('amazon_products.csv')

print('Dataset shape :', df.shape)
print('Columns       :',
list(df.columns))
```

1. Dataset Overview

Problem	Load the dataset and find the total number of products listed.

Solution	`import pandas as pd`
	`import numpy as np`
	
	`df = pd.read_csv('amazon_products.csv')`
	
	`# Count total rows - each row is one product`
	`total_products = df.shape[0]`
	`print('Total Products listed:', total_products)`

2. Category Distribution

Problem	Identify the top 5 product categories with the most listings.
Solution	`top_categories = df['category'].value_counts().head(5)` `print('Top 5 Categories:\n', top_categories)`

3. Pricing Analysis

Problem	Calculate the average discount percentage offered across all products.
Solution	`avg_discount = df['discount_percentage'].mean()` `print(f'Average Discount: {avg_discount}%')`

4. Highest Rated Products

Problem	Find the product(s) with a perfect 5.0 rating and more than 1000 reviews.
Solution	`high_rated = df[(df['rating'] == 5.0) & (df['rating_count'] > 1000)]` `print(high_rated[['product_name', 'rating']])`

5. Correlation: Price vs Rating

Problem	Analyze if expensive products receive higher ratings.
Solution	`correlation = df[['discounted_price', 'rating']].corr()` `print('Correlation Matrix:\n', correlation)`

6. Most Reviewed Category

Problem	Determine which category has the highest total number of user reviews.
Solution	`reviews_by_cat = df.groupby('category')['rating_count'].sum().sort_values(ascending=False)` `print('Most Reviewed Category:', reviews_by_cat.idxmax())`

7. Revenue Potential Analysis

Problem	Identify top 3 products with highest 'Price * Rating Count'.

Solution	`df['popularity_score'] = df['discounted_price'] * df['rating_count']`
	`top_revenue = df.nlargest(3, 'popularity_score')`
	`print(top_revenue[['product_name', 'popularity_score']])`

8. Discount Outliers

Problem	Identify products being sold at a discount greater than 90%.
Solution	`heavy_discounts = df[df['discount_percentage'] > 90]` `print('Products with >90% Discount:\n', heavy_discounts)`

9. Brand Performance

Problem	Find the average rating for 'Samsung' products.
Solution	`samsung_avg = df[df['product_name'].str.contains('Samsung', case=False)]['rating'].mean()` `print('Samsung Average Rating:', samsung_avg)`

10. Price Range Categorization

Problem: Categorize products into Budget, Mid-Range, and Premium.

Solution:

```
def categorize_price(price):
    if price < 1000: return 'Budget'
    elif price < 5000: return 'Mid-Range'
    else: return 'Premium'

df['Price_Bracket'] = df['discounted_price'].apply(categorize_price)
print(df['Price_Bracket'].value_counts())
```